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Advanced Subnetting



- For devices in different networks to communicate, they require a logical address
- IP is one of the network layer protocols used to provide logical addressing
- An IP address is a numeric identifier assigned to each node/host on an IP network
 - ❖ It designates the specific location of a device on the network



Classful and Classless IP

Addressing

- Classful addressing is a historic method assigning IP version 4 addresses based on address classes:
 - i. Class A - for large multinationals
 - ii. Class B –for medium sized companies
 - iii. Class C – for small companies
 - iv. Class D and E are reserved address classes
- Today classless IP addressing is used
- Classless IP is a method assigning IP version 4 addresses without regard for address classes
 - ❖ A network prefix is assigned together with the IP address as it determines how many IPs are available in the given address space

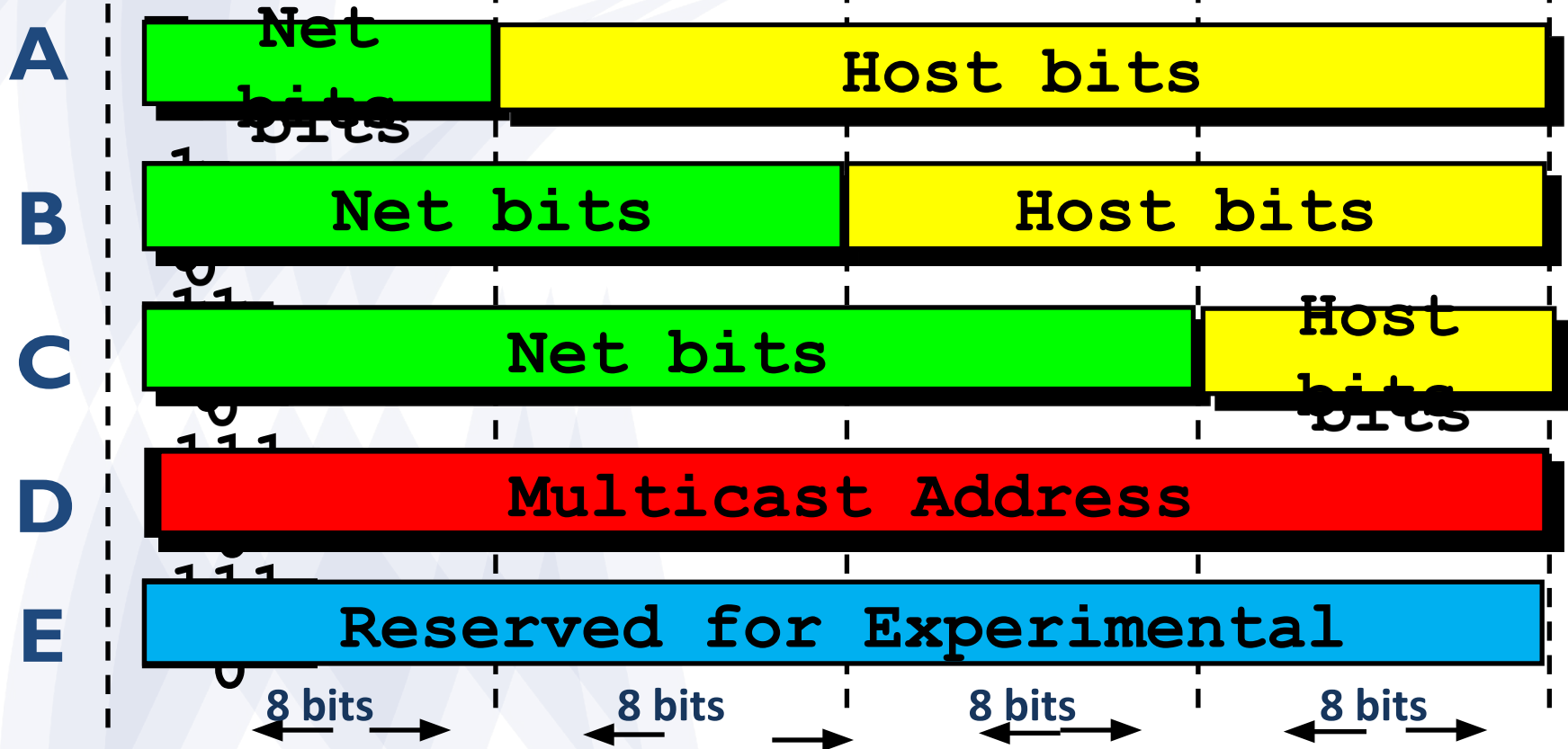
Recap (Computer Networks)

IPv4 Address Classes



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Class



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IPv4 Address Classes...cont



- Summary of IPv4 Addresses

Class	First Octet Range	High order bits (first octet)	Network and Host Parts of the address	No of possible networks	No of hosts per network
A	0-127	0	N.H.H.H	$2^7 = 128$	$(2^{24})-2 = 16,777,214$
B	128-191	10	N.N.H.H	$2^{14}=16,384$	$(2^{16})-2 = 65,534$
C	192-223	110	N.N.N.H	$2^{21}=2,097,150$	$(2^8)-2 = 254$
D	224-239	1110	Reserved for Multicast Communication		
E	240-255	1111	Reserved for Experimental Purposes		

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Special IPv4 Addresses



- The following addresses are not assigned to hosts on a network:
 - i. 0.0.0.0 - reserved for the default route
 - ii. 127.0.0.0 - reserved for loopback testing/local host
 - iii. 169.254.0.0/16 – Link local IPv4 address
 - iv. Test-NET IPv4 address - 192.0.2.0/24 (Test-NET 1) 198.51.100.0/24 (Test-NET2) and 203.0.113.0/24 (Test-NET3)
 - v. 255.255.255.255 - reserved for broadcast communication
 - vi. 224 -239 (address range) - reserved for multicast communication
 - vii. 240-254 (address range) - reserved for experimental purposes

Networks)

Public and Private Addresses



- Public addresses
 - ❖ Also known as **registered numbers**
 - ❖ Routed over the internet
 - ❖ They are assigned by Regional Internet Registries (RIRs) e.g. AfriNic manages IPs for the African region
 - ❖ Other regional internet registries include: RIPE (Europe, Middle East & Central Asia), ARIN (North America & Canada), LACNIC (Latin America & the Caribbean), APNIC (Asia and Pacific Rim)
 - ❖ IANA – manages IP addresses globally

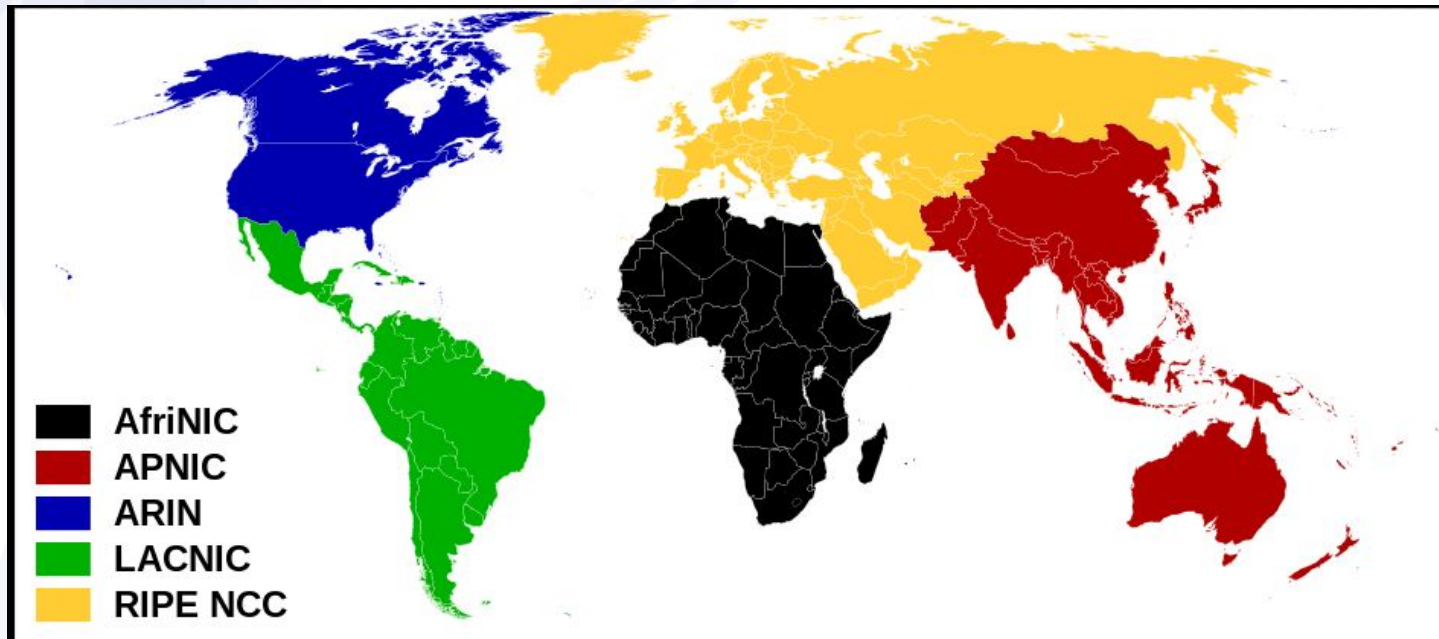
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Public and Private Addresses



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Regional Internet Registries (RIRs)



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Public and Private Addresses



Private addresses

- ❖ Also known as **unregistered numbers**, chosen by the network administrator
- ❖ Used within private networks, not routed over the internet

Private IP Ranges

10.0.0.0 – 10.255.255.255 or 10.0.0.0/8

172.16.0.0 - 172.31.255.255 or 172.16.0.0/12

192.168.0.0 - 192.168.255.255 or 192.168.0.0/16

- ❖ **NB:** Apart from these private IPs and all other mentioned special IPs we can assume that all the remaining IP addresses are public IPS

Recap (Computer Networks)

Types of Addresses found on a Network



1. **Network address / network ID** – uniquely identifies or groups together hosts on the same network or subnet
2. **Host address** – uniquely identifies a device/host on a network
3. **Broadcast address** – used to send a message to all devices on the same network segment/ subnet

NB: In Binary:

- The network /subnet address has all zeros in the host portion
- A broadcast address has all ones in the host portion of the address

Types of Addresses

found on a Network

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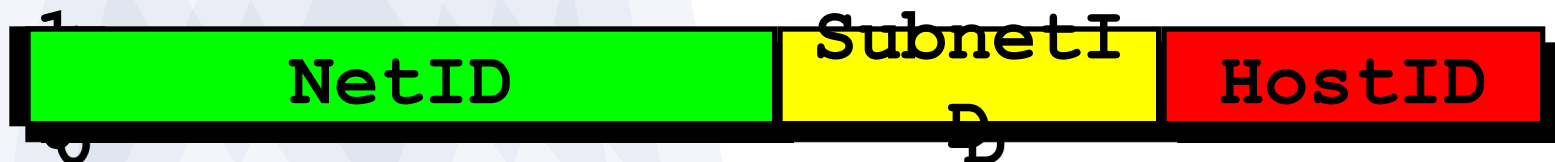
- The network /subnet address has all zeros in the host portion
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Recap (Computer Networks) Subnetting



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- Is the process of sub-dividing a given host address space into logical groupings called sub-networks (subnets)
- Achieved through borrowing bits from the host portion of an address and assigning them to the network portion, so that the network bits increase



Recap (Computer Networks)



Why do we subnet?

1. **Security** - easier to manage the network's security when subnets are used. E.g. access to certain parts or resources on the network can be restricted based on subnets
2. **Address Management** – a network can easily scale up or down, based on subnets and the assignable IP addresses available easily established
3. **Improve network performance** – with subnets management of network traffic is made easier e.g. network broadcasts can easily be confined to a given subnet or even a virtual LAN (V-LAN)
4. **Fault isolation** – It is easier to isolate network faults within subnets as opposed to one large network

Formula

To determine how many subnets an address can have

2^n Total number of subnets

Where n =number of borrowed bits

To determine number of hosts an address can have per subnet

2^h Total number of hosts per subnet

$2^h - 2$ Number of usable hosts per subnet

*Where h =number of remaining bits
(unborrowed bits)*

Basic Rules for

Subnetting

1. Always borrow bits from the **host portion only** (*if using a classful address*)
2. Always borrow bits from the **leftmost host bit** progressing toward the right hand side
3. Never **borrow all bits in the last octet** (*you can only borrow up to the 30th bit, i.e. at least 2 bits must remain*)
4. The least number of bits you can borrow is 2 (**if you are using legacy equipment**)
5. Whenever subnets are created the subnet mask changes, that is, the length of the mask increases, since the number of network bits is increased to create the subnets

Basic Subnetting

Example (*From Takeaway Assignment 1*)

Task One

You have been assigned the address: 192.168.25.0/24.
Subnet this address such that each subnet has 25 hosts per subnet.

- a) How many bits will you borrow?
- b) How many bits will remain in the address?
- c) How many subnets will you have in total?
- d) How many usable host addresses will you have per subnet?
- e) Place your results in the subnet chart (*ensure that you show your working*)
- b) What is the new subnet mask for the subnets in this network in
 - (i). binary notation (ii). decimal notation
 - (iii) prefix notation

Recap (Computer Networks)

Basic Subnetting



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Task Two

Sketch a simple logical network topology to show how any TWO of your subnets could be connected on a single router. Assume that each subnet will be placed on its own router interface. Since each subnet represents a LAN and each LAN has a number of devices, include just ONE end device (e.g. PC) to represent those on each subnet. On each router port indicate the IP address that you would configure and its subnet mask in prefix form. On each end device indicate the IP address, prefix and the default gateway that you would configure.



- A **subnet mask** is a number used to identify the network and host portions of an assigned logical address
- IPv4 masks are 32 bits long (4 octets)
- Network portion of the mask is represented by all ones while the host portion is represented by all zeros
- A bitwise AND between a given IP and its subnet mask results in the network ID to which the IP address belongs



IPv4 masks can be represented in:

- a) Dotted decimal notation e.g. 255.0.0.0
- b) Dotted binary notation e.g.
11111111.00000000.00000000.00000000
- c) Prefix notation or slash format e.g. /8

- To determine the network/subnetwork to which an address belongs, devices such as routers perform a bit-wise AND process between the IP address and its corresponding subnet mask

Rules for ANDing

$$1 \text{ AND } 1 = 1$$

$$1 \text{ AND } 0 = 0$$

$$0 \text{ AND } 1 = 0$$

$$0 \text{ AND } 0 = 0$$

Example:

A host has the IP 192.168.64.3 and a subnet mask of 255.255.255.224. What network does this host address belong to?

IP Address----- 11000000.0101000.01000000.00000011

Subnet Mask--- 11111111.11111111.11111111.11100000

AND result----- 11000000.10101000.01000000.00000000

192. 168 . 64 . 0

- The address **192.168.64.3** belongs to the subnet **192.168.64.0**

Host Address Allocation

Once subnets have been determined together with the host address ranges the actual configuration on the hosts can be done either:

- a) Statically i.e. manually configuring the IP addresses
- b) Dynamically using protocols such as DHCP

NB: You can only configure/use addresses in the usable range of your subnet chart.

Networks)

Host Address Configuration



At minimum a network interface should be configured with:

- a. An IP Address (only usable host addresses are assignable)
- b. A subnet mask
- c. Default gateways (*only where the device requires one e.g. on a PC a default gateway may be necessary*)

Subnetting for Large Network Requirements



- Default class B address format – **N.N.H.H**
 - There are 16 host bits to borrow from (i.e. H.H)
- Default class A address format – **N.H.H.H**
 - There are 24 host bits to borrow from (i.e. H.H.H)
- The general rule is to start borrowing from the left most bit closest to the network portion of the address progressing toward the right
- NB:
 - The general formulae still apply
 - Since 2^h will sometimes yield a value bigger than 255, which is higher than the allowable in an octet, you need to know where the incremental is done to produce the correct values for your subnet chart

Subnetting for Large Network Requirements



Example:

A network administrator has the IP: 172.16.0.0/16.

Assume that the administrator will subnet this address so that he/she has subnets with 1000 host addresses in each subnet.

1. How many
 - a) Bits will remain?
 - b) Bits will be borrowed?
 - c) Usable host addresses will there be per subnet?
 - d) Subnets will be created?
2. Write a subnet chart to represent the first 5 and the last 2 subnets
3. Sketch a simple topology showing connection for users in subnet 2. You will use one router, one switch, two user-devices. Indicate the IP settings that you will configure.

Subnetting for Large Network Requirements



Determining the correct incremental value

- **Method 1**

- **Rule:** If the number of remaining bits (*'unborrowed' bits*) is **equal to 8 or above 8 but less than 15 then:**

- $2^h \div 256$

- Examples: $2^9 \div 256 = 2$

- $2^8 \div 256 = 1$

- $2^{13} \div 256 = 32$

- $2^{10} \div 256 = 4$

- Your incremental value will therefore be equal to the result of the calculation as shown above

NB: *You will increment in the octet where some bits were borrowed and some remained by the computed value above*

Subnetting for Large Network Requirements



Writing the Correct Subnet Chart

- **Example:** IP Address is 172.16.0.0/16 and h=10
- $2^{10} \div 256 = 4$
- Part of the subnet chart

Sub No	Subnet Address	Usable Host Addresses	Broadcast Address
0	172.16.0.0	172.16.0.1-172.16.3.254	172.16.3.255
1	172.16.4.0	172.16.4.1-172.16.7.254	172.16.7.255
2	172.16.8.0	172.16.8.1-172.16.11.254	172.16.11.255

In this example, the third octet has some bits borrowed and some remaining hence the computed value is added here

Subnetting for Large Network Requirements



Determining the correct incremental value

- **Method 2**

- **Rule:** If the number of remaining bits is **equal to 8 or above 8** then your incremental value is given by:
 - 2^h
 - *where h = the number of remaining ('unborrowed' bits) bits in the octet in which some bits were borrowed.*

Example: If $h = 10$

- then the bits in the class B will look like this:
N.N.00000000.00000000
- where **N.N** are the network bits of the default class B address , **000000** are the borrowed bits and **00.00000000** are the 10 (ten) remaining bits in this class B address
 - Therefore the incremental value will be given by **2^2** since only 2 bits remained in the third octet

(See subnet chart on next page)

Subnetting for Large Network Requirements



Writing the Correct Subnet Chart

- **Example:** IP Address is 172.16.0.0/16 and h=10

N.N.00000000.00000000

$$2^2 = 4$$

- Part of the subnet chart

Sub No	Subnet Address	Usable Host Addresses	Broadcast Address
0	172.16.0.0	172.16.0.1-172.16.3.254	172.16.3.255
1	172.16.4.0	172.16.4.1-172.16.7.254	172.16.7.255
2	172.16.8.0	172.16.8.1-172.16.11.254	172.16.11.255

NB: in this example: **The broadcast address** = next subnet address deduct 1 in the third octet, and place 255 in the last octet. The usable addresses follow the regular rules you had learnt.

Subnetting for Large Network Requirements



Determining the correct incremental value

- **Method 2**

- **Rule:** Find the magic number. The magic number is given by the position of the last “1” in the new subnet mask (in binary)

Example: If $h = 10$

- then the bits in the class B will look like this: **N.N.00000000.00000000**
- New subnet mask in binary will look like this:
11111111.11111111.1111100.00000000
- where **11111111.11111111** are the network bits of the default class B address , **11111** are the borrowed bits which now form part of the network bits and **00.00000000** are the 10 (ten) remaining bits in this class B address
- **1** is the last 1 in the mask
 - Incremental is therefore given by 2^2 since this **1** is in position 2 of the binary number

(See subnet chart on next page)

Subnetting for Large Network Requirements



Writing the Correct Subnet Chart

- **Example:** IP Address is 172.16.0.0/16 and h=10

11111111.11111111.11111100.00000000

$$2^2 = 4$$

- Part of the subnet chart

Sub No	Subnet Address	Usable Host Addresses	Broadcast Address
0	172.16.0.0	172.16.0.1-172.16.3.254	172.16.3.255
1	172.16.4.0	172.16.4.1-172.16.7.254	172.16.7.255
2	172.16.8.0	172.16.8.1-172.16.11.254	172.16.11.255

NB: in this example: **The broadcast address** = next subnet address deduct 1 in the third octet, and place 255 in the last octet. The usable addresses follow the regular rules you had learnt.



Variable Length Subnet Masks (VLSM)

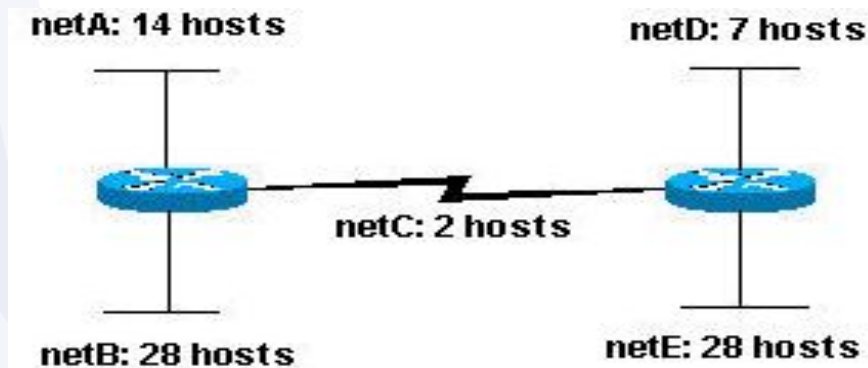
Variable Length Subnet Masking



- Variable Length Subnet Masking (VLSM) – allows division of address space based on the size of networks, thereby using address space efficiently
- It is achieved by simply subnetting a subnet
 - ❖ Subnets can therefore bear different/varying subnet masks
- Rationale behind VLSM is to conserve address space by allocating just enough addresses to any given subnetwork
- It also helps in planning networks for future expansion

Example

- Assume the network address: 204.15.5.0/24. Subnet the address in order to create the network in with the host requirements shown.
 - a. Subnet the address without regard for network size. Place your results in a subnet chart.
 - b. Subnet the address with regard for network size, ensuring that your subnets are not wasteful. Place your results in a subnet chart.
 - c. Comment on the two charts



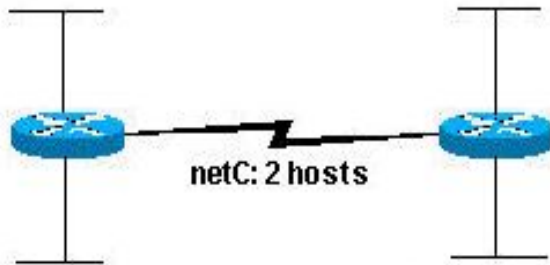
Solution



Subnet No	Network	Subnet Address	Usable Host Address Range	Broadcast Address
0	Net B	204.15.5.0	204.15.5.1-204.15.5.31	204.15.5.31
1	Net E	204.15.5.32	204.15.5.33-204.15.5.62	204.15.5.63
2	Net A	204.15.5.64	204.15.5.65-204.15.5.94	204.15.5.95
3	Net D	204.15.5.96	204.15.5.97-204.15.5.126	204.15.5.127
4	Net C	204.15.5.128	204.15.5.129-204.15.5.158	204.15.5.159
5		204.15.5.160	204.15.5.161-204.15.5.190	204.15.5.191
6		204.15.5.192	204.15.5.193-204.15.5.222	204.15.5.223
7		204.15.5.224	204.15.5.225-204.15.5.254	204.15.5.255

netA: 14 hosts

netD: 7 hosts



netB: 28 hosts

netE: 28 hosts

Without Using VLSM

- ❑ You subnet for the largest subnet (network) only
 - i. Ensure that the total number of subnets is sufficient to fit all the required subnets. If not sufficient you will need a different address. **NB: In this scenario, each network (A to E) occupies an entire subnet**
 - ii. The new subnet mask is the same for all subnets. In this example: it will be 255.255.255.224 or /27
 - iii. It is wasteful where the number of usable host addresses ends up being much larger than required e.g. Net C requires only 2 usable IPs but has 30 as per the chart below

Solution

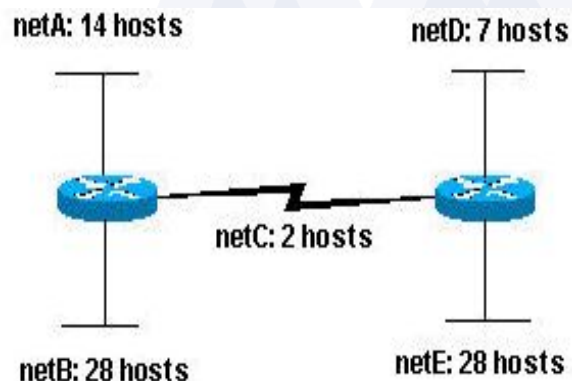


Subnet No	Network	Subnet Address	Usable Host Address Range	Broadcast Address	Prefix	Subnet Mask in Decimal format
0	Net B	204.15.5.0	204.15.5.1-204.15.5.31	204.15.5.31	/27	255.255.255.224
1	Net E	204.15.5.32	204.15.5.33-204.15.5.62	204.15.5.63	/27	255.255.255.224
2	Net A	204.15.5.64	204.15.5.65-204.15.5.78	204.15.5.79	/28	255.255.255.240
3	Net D	204.15.5.80	204.15.5.81-204.15.5.94	204.15.5.95	/28	255.255.255.240
4	Net C	204.15.5.96	204.15.5.97-204.15.5.98	204.15.5.99	/30	255.255.255.252
5		204.15.5.100	204.15.5.101-204.15.5.102	204.15.5.103	/30	255.255.255.252
6		204.15.5.104	204.15.5.105-204.15.5.106	204.15.5.107	/30	255.255.255.252
7		204.15.5.108	204.15.5.109-204.15.5.110	204.15.5.111	/30	255.255.255.252
8		204.15.5.112	204.15.5.113-204.15.5.114	204.15.5.115	/30	255.255.255.252

Using VLSM

- You determine the borrowed and remaining bits for each subnet
- Start from the largest subnet and work your way down to the smallest subnet
- The new subnet mask may not be the same for some subnets since the number of borrowed and remaining bits vary per subnet.
- It is not wasteful e.g. Net C requires only 2 usable IPs but only 2, Net D requires 7 and has 14, Net A requires 14 and has exactly 14 usable IPs

In this solution: the subnets from Net C all bear the same number of addresses hence same mask / prefix. We can

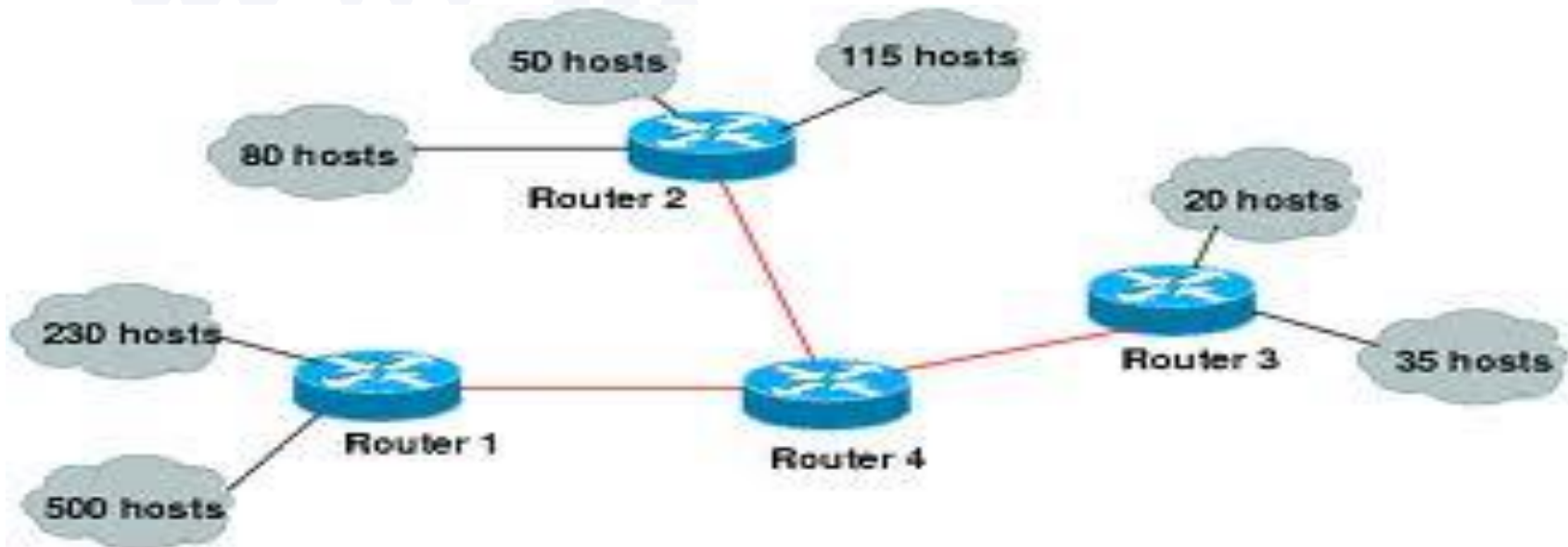


Question



- A network administrator intends to use the address 172.17.0.0/16 to create subnets for the private networks on routers 1, 2, 3 and 4 shown in the topology above. Show how his subnet chart will look like if he subnets the address:
 - Without varying the network sizes
 - If he varies the network sizes accordingly

(NB: for both questions write the first 10 subnets and last 2 only)



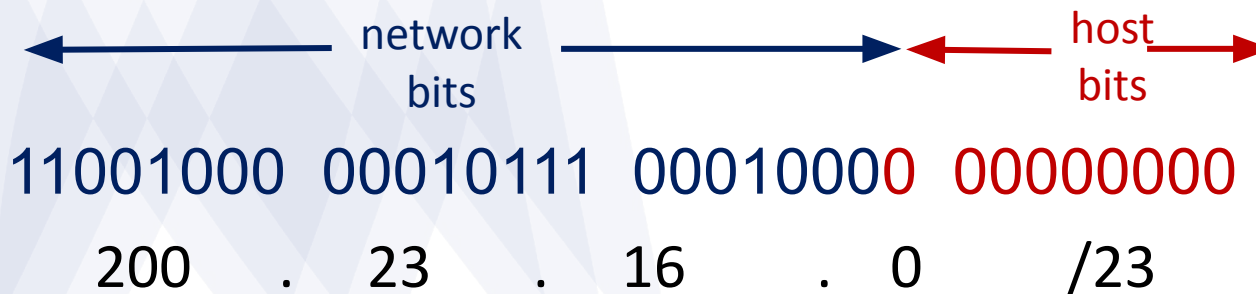


Classless Inter-domain Routing (CIDR)

Classless Inter-Domain Routing (CIDR)



- Is a way to allocate and specify the Internet Protocol addresses used in inter-domain routing more flexibly than with the original system of Internet Protocol(IP) address classes
- In CIDR, class does not matter, instead when network address is issued the length of the network prefix **must be** specified as part of the IP address
 - ❖ the prefix length varies depending on the number of host bits that are needed e.g. 200.23.16.0/23



Classless Inter-Domain Routing (CIDR)



- In CIDR each IP address has a *network prefix* that identifies either an **aggregation of network gateways** or **an individual gateway**
 - ❖ **For example:** 192.168.0.0/22 aggregates 4 class C networks into 1.
- **Note:**
 - ❖ CIDR uses VLSM to allocate addresses to networks/subnetworks according to size rather than class
 - ❖ CIDR addresses can be divided further (subnetted) depending on need
 - ❖ Allows aggregation of networks with **masks smaller** than those of the default of the address class e.g. a /22 mask can be assigned to a class C address whose default mask based on the class would have been /24

Route Summarization (Aggregation)



- Route summarization is the aggregation of various contiguous network addresses into one summary address also known as a **summary route** or an **aggregate route**.
- **Example:** the networks

172.20.0.0
172.21.0.0
172.22.0.0
172.23.0.0



can be aggregated into **172.20.0.0/14**

Route Summarization (Aggregation)



- Route summarization used in CIDR allows for aggregation of networks with **prefixes smaller** than those of the default of the address class
- When an aggregate or summary route has a prefix smaller than that of the default of the class it can be referred to as a **supernet**
- CIDR is therefore is sometimes referred to as **supernetting**
- **NB:**
 - ❖ Route aggregation / summarization does not always result in supernets

How to Summarize/Aggregate Routes



- Therefore the summary route is : **172.20.0.0/14**
- 172.20.0.0/14 is a **supernet route** since the prefix is smaller than that of a class B address
- ◆ **NB:** Remember 172 is a class B address with a default prefix of 16.

Calculating a Route Summary

Step 1: List networks in binary format.

172.20.0.0	10101100	. 00010100	. 00000000	. 00000000
172.21.0.0	10101100	. 00010101	. 00000000	. 00000000
172.22.0.0	10101100	. 00010110	. 00000000	. 00000000
172.23.0.0	10101100	. 00010111	. 00000000	. 00000000

Step 2: Count the number of left-most matching bits to determine the mask. 14 matching bits, /14 or 255.252.0.0

Step 3: Copy the matching bits and add zero bits to determine the network address.

172.20.0.0	10101100	. 00010100	. 00000000	. 00000000
------------	----------	------------	------------	------------

Copy

Add zero bits

Benefits of CIDR



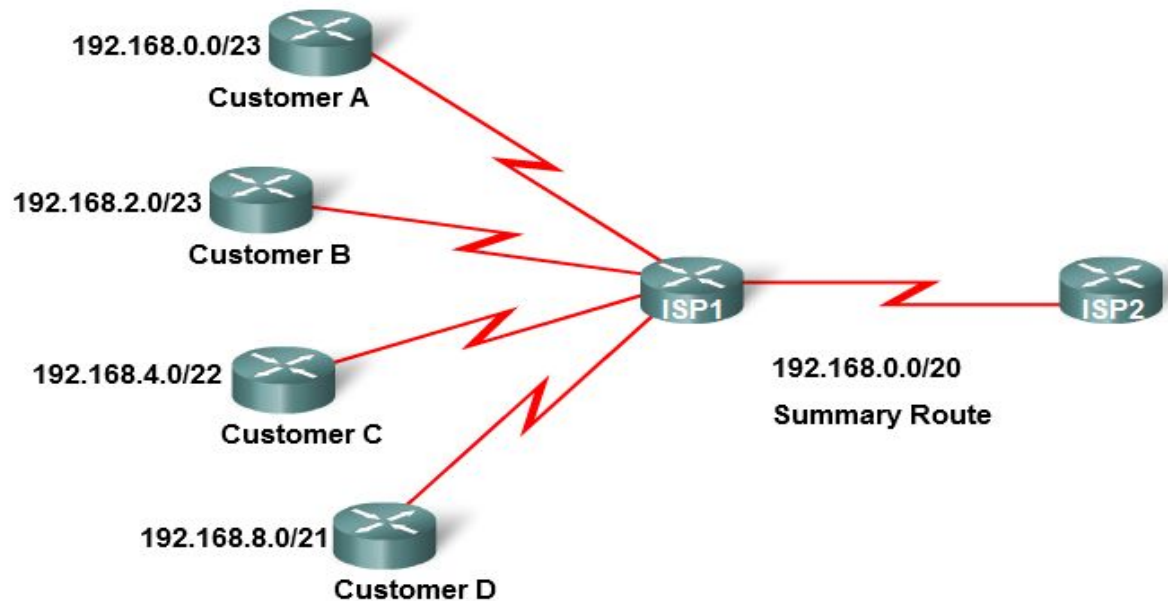
1. Allows us to have a larger address space hence more addresses to assign since the class does not matter
 - ❖ The bit boundary of address classes does not determine the number of hosts assignable in a given block.
 - ❖ E.g. the CIDR address 192.168.0.0/23 allows for 512 total host addresses which would be unlikely if the IP address was treated as a normal class C address
2. Efficient use of address space
 - ❖ CIDR uses variable length subnet masks (VLSM) thus it ensures that networks have just the right number of usable host addresses

Benefits of CIDR



3. Allows us to summarize many network addresses into a single address (aggregate route) which helps in reducing the size of the routing table hence:
- ❖ Reduced router memory requirements and
 - ❖ Reduced number of CPU cycles for routing processes
 - ❖ Faster route resolution (due to fewer routing table entries)

In the diagram the routing table of ISP 2 will reduce in size significantly when the summary route is used to reach the customers' networks



CIDR – Challenge Question



An organization requires 4000 addresses for the devices on its network. Assume that the internet authorities would like to assign the organization a CIDR address in the class C range:

- a. How many class C addresses would need to be bundled together in the CIDR address
- b. Choose an address for this scenario and show what the network address would be.



- Data Communications and Networking by Forouzan
 - <https://www.countryipblocks.net/how-ipv4-network-assignments-work>)
 - <https://tools.ietf.org/html/rfc1918>
 - <https://tools.ietf.org/html/rfc5735>
 - ISoc -
<http://www.internetsociety.org/deploy360/resources/ipv6-training-from-boson-com/>
 - IPv6 for Dummies (2012) -
www.xnetworks.es/contents/Infoblox/IPv6forDummies.pdf